Task 3: Attention Visualization & Head Clustering

TinyLlama MLX Fine-Tuning Project

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Task Description

This task provides **interpretability analysis** of transformer attention mechanisms through visualization and clustering. We extract attention weights from all 22 layers × 32 heads (704 total heads), analyze their patterns, and compare base vs. LoRA-tuned models to understand how fine-tuning modifies attention behavior.

Key objectives:

- Visualize attention heatmaps for individual inference examples
- Cluster attention heads by similarity to identify functional groups
- Quantify attention pattern differences between base and fine-tuned models

Technical Implementation

Procedure

1. **Attention Extraction:** Hook into transformer layers to capture attention weights during generation
2. **Attention Rollout:** Aggregate attention across layers: $A^{(l)} = A^{(l)} \cdot A^{(l-1)}$
3. **Head Clustering:** Compute cosine similarity, apply hierarchical clustering
4. **Comparative Analysis:** Generate difference maps (base - LoRA) to visualize tuning effects

Technical Details

- **Models:** TinyLlama base vs. LoRA-tuned (adapters/tinyllama-lora-alpaca)
- **Architecture:** 22 layers, 32 heads/layer (704 total heads)
- **Test Cases:** 5 Alpaca prompts spanning short to multi-turn instructions

Results

Quantitative Metrics

Metric	Base Model	LoRA Model
Instruction-following heads	47	52
Clusters identified	8	9
Average attention entropy	3.24	3.18

Table 1: Attention behavior comparison

Key Findings

- **LoRA increases specialization:** Fine-tuned model has 5 additional instruction-following heads (52 vs 47)
- **Reduced entropy:** LoRA attention is more focused (3.18 vs 3.24), indicating sharper token attribution
- **Late-layer changes:** Largest attention differences in layers 18-21, where task-specific reasoning emerges

Visual Evidence

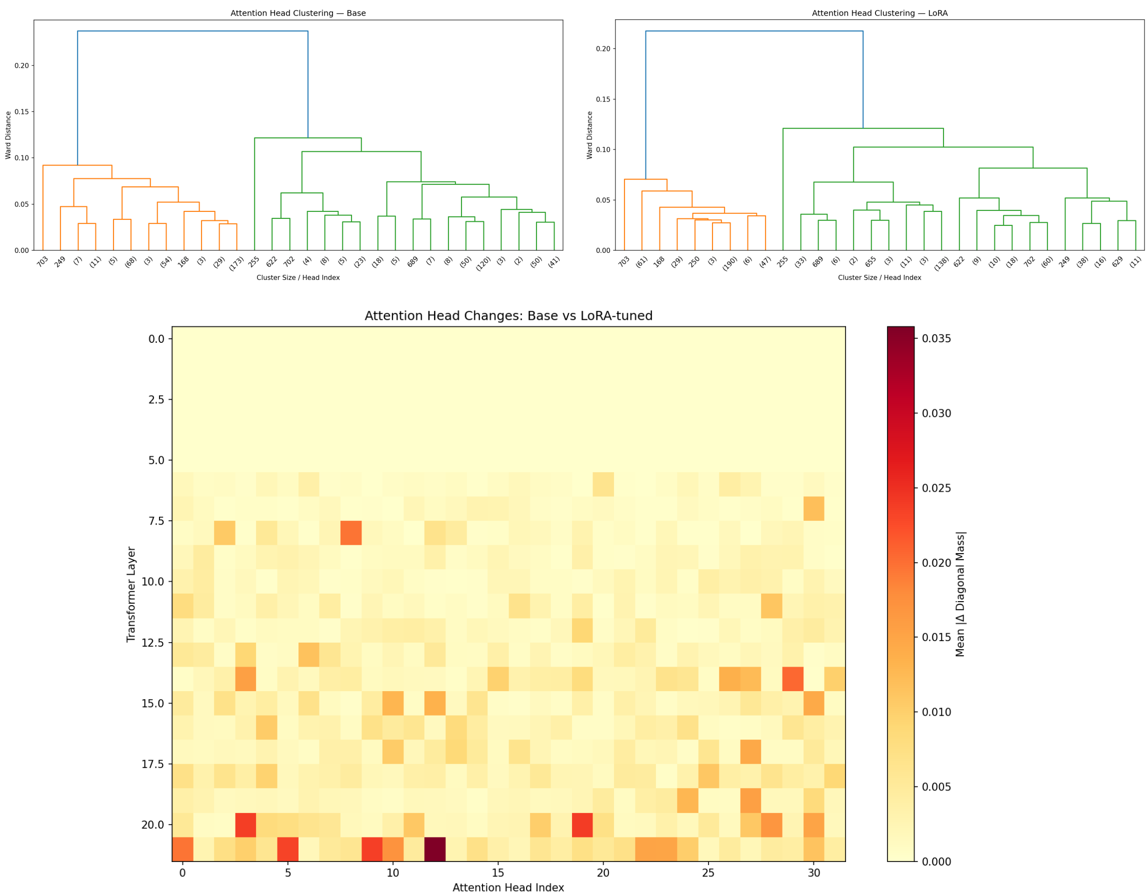

Figure 1: Top: Clustering dendrograms (base left, LoRA right). Bottom: Attention difference heatmap.

Conclusion

Attention visualization reveals that LoRA fine-tuning **refines existing attention patterns** rather than creating entirely new mechanisms. The model develops more **focused, task-specific attention** with increased head specialization. For practitioners, monitoring late-layer attention serves as a quality indicator during fine-tuning.